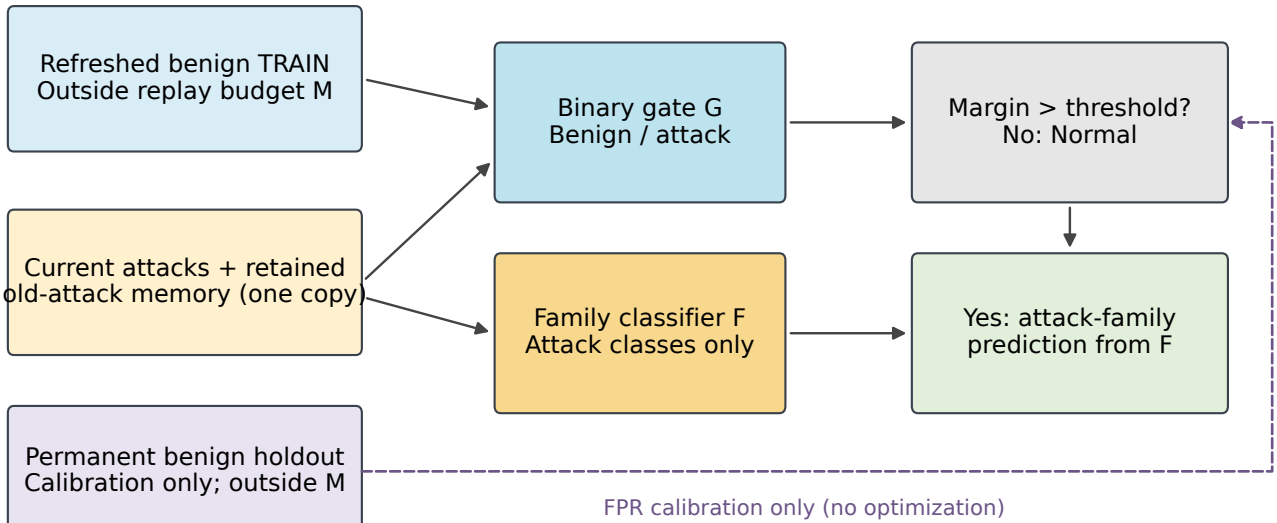
Refreshed benign TRAIN
Outside replay budget M
Current attacks + retained
old-attack memory (one copy)
Permanent benign holdout
Calibration only; outside M
Binary gate G
Benign / attack
Family classifier F
Attack classes only
Margin > threshold?
No: Normal
Yes: attack-family
prediction from F
FPR calibration only (no optimization)